

Detection and Response to a Real SSH Brute Force Attack Using Wazuh, TheHive, and MISP

End-to-End SOC Incident Handling: Detection, Investigation, Mitigation, and Threat Intelligence Sharing

INTRODUCTION

This project documents a **real-world SSH brute force attack** detected and handled in a live SOC-style environment using **Wazuh SIEM**, **TheHive case management**, and **MISP threat intelligence sharing**.

The incident involved repeated unauthorized SSH login attempts originating from an **unknown external IP address located in the United States**, while the target system was hosted and monitored from **Bangalore, India**.

The attack was detected in real time, investigated, mitigated, documented, and shared with the wider threat intelligence community.

This report follows the **complete SOC incident response lifecycle** from detection to closure.

TOOLS & TECHNOLOGIES USED

- **Wazuh SIEM** – Detection and log analysis
- **TheHive 5** – Alert triage, case management, investigation workflow
- **MISP** – Threat intelligence enrichment and IOC sharing
- **Slack** – Real-time alert notification
- **Linux (Ubuntu)** – Target system
- **Fail2Ban** – Brute force mitigation
- **iptables** – Network-level blocking
- **MITRE ATT&CK Framework** – Threat classification

INCIDENT OVERVIEW

- **Attack Type:** SSH Brute Force
- **MITRE Technique:** T1110 – Brute Force
- **Target:** Linux host monitored by Wazuh agent
- **Source IP:** External public IP (United States)
- **Detection Method:** Wazuh correlation rules
- **Severity:** Medium
- **Outcome:** True Positive

INCIDENT TIMELINE

TIME (UTC)	ACTION
00:48	Slack alert received
00:50	Brute force activity started
00:52	Last observed attempt
00:53	Alert triaged in TheHive
01:00	Investigation confirmed in Wazuh
01:10	Case created and documented
01:30	Mitigation actions applied
01:50	Case Closed
02:10	IOC shared back to MISP

◆ Step 1: Alert Notification Received (Slack)

A real-time alert notification was received via Slack indicating **SSH Brute Force Detection** triggered by Wazuh.

The alert included severity level, detection time window, and the attacking IP address.

This confirmed that automated detection mechanisms were functioning correctly.



Wazuh-SOC-Alerts APP 00:48

Alerting Notification action

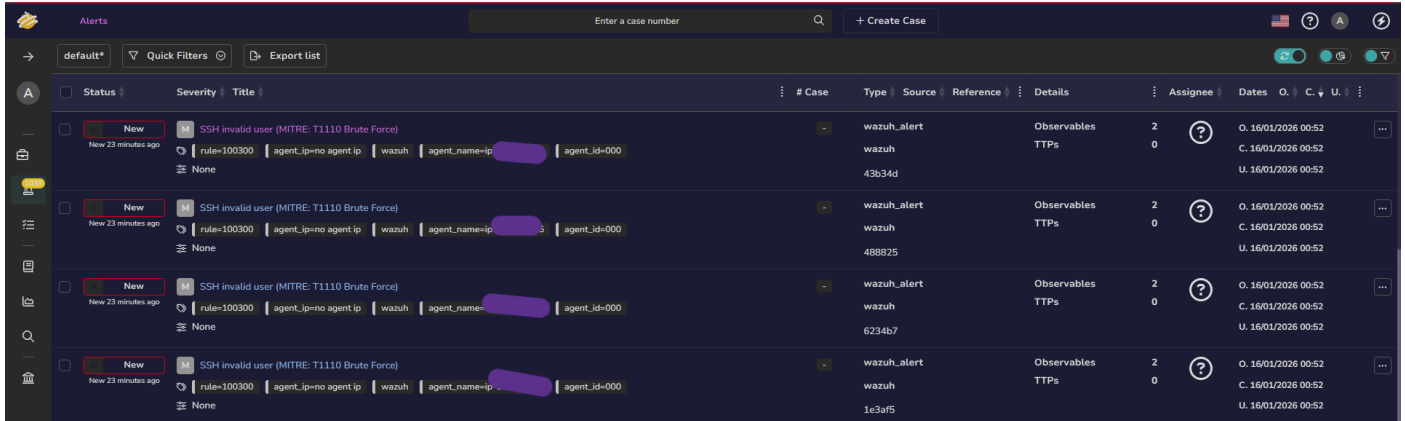
Monitor SSH Brute Force Detection just entered alert status. Please investigate the issue.

- Trigger: SSH Brute Force Alert
- Severity: 2
- Period start: 2026-01-15T19:17:48.944Z UTC
- Period end: 2026-01-15T19:18:48.944Z UTC
- Deduped Alerts:
- New Alerts:
 - * p3MYw5sBOxFRVNsKnTij : 102.208.186.12
- Completed Alerts:



◆ Step 2: Alert Observed in TheHive

After receiving the Slack notification, the alert was immediately located in **TheHive Alerts dashboard**. The alert title indicated “**SSH invalid user – Brute Force (MITRE T1110)**”, confirming credential attack behavior.



The screenshot shows the TheHive Alerts dashboard with a list of four alerts. Each alert is titled 'SSH invalid user (MITRE: T1110 Brute Force)' and is marked as 'New' (New 23 minutes ago). The alerts are filtered by 'rule=100300', 'agent_ip=no agent ip', 'wazuh', and 'agent_name=ip'. The table columns include Status, Severity, Title, # Case, Type, Source, Reference, Details, Assignee, Dates, and a menu icon. The alerts are assigned to 'wazuh' and have a status of 'New'.

Status	Severity	Title	# Case	Type	Source	Reference	Details	Assignee	Dates
New	High	SSH invalid user (MITRE: T1110 Brute Force)	43b34d	wazuh_alert	wazuh	43b34d	Observables TTPs	2 0	O. 16/01/2026 00:52 C. 16/01/2026 00:52 U. 16/01/2026 00:52
New	High	SSH invalid user (MITRE: T1110 Brute Force)	488825	wazuh_alert	wazuh	488825	Observables TTPs	2 0	O. 16/01/2026 00:52 C. 16/01/2026 00:52 U. 16/01/2026 00:52
New	High	SSH invalid user (MITRE: T1110 Brute Force)	6234b7	wazuh_alert	wazuh	6234b7	Observables TTPs	2 0	O. 16/01/2026 00:52 C. 16/01/2026 00:52 U. 16/01/2026 00:52
New	High	SSH invalid user (MITRE: T1110 Brute Force)	1e3af5	wazuh_alert	wazuh	1e3af5	Observables TTPs	2 0	O. 16/01/2026 00:52 C. 16/01/2026 00:52 U. 16/01/2026 00:52

◆ Step 3: Alert Status Updated and Assigned

The alert status was changed to **In Progress**, and the case was assigned to myself. An initial triage summary was added to document that investigation had begun.



The screenshot shows the 'Change the alert status' dialog in TheHive. The status is set to 'In Progress'. A summary is added: 'Initial triage started for SSH brute force alert generated by Wazuh. Alert indicates multiple failed SSH authentication attempts against the host. Proceeding with validation using SIEM logs and threat analysis.' The assignee is 'abdulrehmanthehivesoc'.

Change the alert status

Status
● In Progress

Summary
Initial triage started for SSH brute force alert generated by Wazuh.
Alert indicates multiple failed SSH authentication attempts against the host.
Proceeding with validation using SIEM logs and threat analysis.

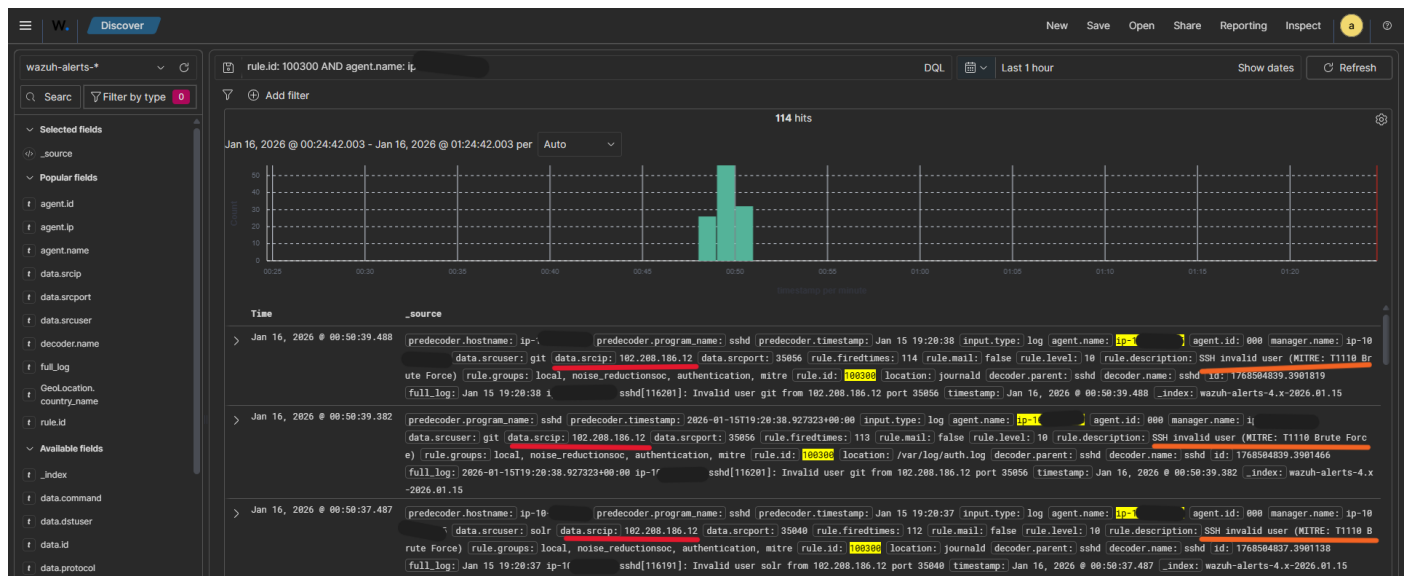
Assignee
A abdulrehmanthehivesoc

◆ Step 4: Investigation in Wazuh SIEM

The alert was validated by pivoting to **Wazuh Discover**.

Multiple failed SSH authentication attempts were observed within a short time window, originating from the same external IP and targeting multiple invalid usernames.

This confirmed malicious brute force behavior.



◆ Step 5: Case Creation in TheHive

A new case titled “**Confirmed SSH Brute Force Attack – Linux Host**” was created.

Severity was set to **Medium**, and TLP was marked as **Amber**.

A detailed description was added summarizing the detection and investigation findings.

The screenshot shows the 'Create case' form in TheHive. The form includes the following fields:

- Title:** Confirmed SSH Brute Force Attack – Linux Host
- Date:** 16/01/2026 01:25
- Severity:** MEDIUM
- TLP:** TLP:AMBER
- PAP:** PAP:AMBER
- Tags:** rule=100300, agent_ip=no agent ip, wazuh, agent_name=ip-10, agent_id=000
- Description:** This case was created following a Wazuh alert indicating SSH brute force activity. During investigation, multiple failed SSH authentication attempts were observed in Wazuh Discover logs. The attack involved repeated login attempts using invalid usernames from a single external IP address. The activity aligns with MITRE ATT&CK technique T1110 (Brute Force). Threat intelligence enrichment and mitigation actions are being performed.

◆ Step 6: Case Documentation and Description

The case description documented:

- Detection source (Wazuh)
- Investigation findings
- Attack pattern
- MITRE ATT&CK mapping

This ensures proper incident traceability.

The screenshot shows the Wazuh dashboard interface for Case #2, titled "Confirmed SSH Brute Force Attack - Linux Host". The case is assigned to "abdulrehmanthehivesoc" and has a status of "New". The description field contains the following text:

This case was created following a Wazuh alert indicating SSH brute force activity. During investigation, multiple failed SSH authentication attempts were observed in Wazuh Discover logs. The attack involved repeated login attempts using invalid usernames from a single external IP address. The activity aligns with MITRE ATT&CK technique T1110 (Brute Force). Threat intelligence enrichment and mitigation actions are being performed.

The timestamp is 2026-01-15T19:20:39.488+0000.

◆ Step 7: Observables Added

The attacker IP address was added as an **observable (ip-src)** and marked as an IOC. Additional contextual information was provided through comments.

The screenshot shows the Wazuh dashboard interface for Case #2, titled "Confirmed SSH Brute Force Attack - Linux Host". The case is assigned to "abdulrehmanthehivesoc" and has a status of "New". The observables section shows two entries:

Flags	Data type	Value/Filename	Dates	S	C	U
☐	hostname	ip-102.1208.1186.112	S. 16/01/2026 01:28			
☐	target-host	102.1208.1186.112	C. 16/01/2026 01:28			

◆ Step 8: MITRE ATT&CK Mapping

The attack was mapped to:

- **T1110 – Brute Force**
- Tactic: Credential Access
- This provides standardized threat classification.

#2 Confirmed SSH Brute Force Attack - Linux Host

id -88383552

Created by abdulrehmanthehivesoc

Created at 16/01/2026 01:27

SEVERITY:MEDIUM

TLP:AMBER PAP:AMBER

Assignee

abdulrehmanthehivesoc

Status

New

General Tasks (0) Observables (2) TTPs (2) Attachments Timeline Report Pages History

default

Export list

Tactic	Technique	By	Dates	O	C	U
credential-access	T1110 - Brute Force	A	O: 16/01/2026 00:51 C: 16/01/2026 01:41			

◆ Step 9: Threat Intelligence Lookup in MISP

The attacker IP address was searched in MISP using the **Search Attributes** feature to check for existing threat intelligence.

No matching attributes or events were found, indicating the IP was not previously reported in MISP.

Home

Event Actions

Dashboard

Galaxies

Input Filters

Global Actions

Sync Actions

Administration

Logs

API

List Events

Add Event

Import from...

REST client

List Attributes

Search Attributes

View Proposals

Events with proposals

View delegation requests

View periodic summary

Export

Automation

Search Attribute

You can search for attributes based on contained expression within the value, event ID, submitting organisation, category and type.

For the value, event ID and organisation, you can enter several search terms by entering each term as a new line. To exclude things from a result.

For string searches (such as searching for an expression, tags, etc) - lookups are simple string matches. If you want a substring match encapsulate with %.

Containing the following expressions

102.208.186.12

Having tag or being an attribute of an event having the tag

Being attributes of the following event IDs, event UUIDs or attribute UUIDs

From the following organisation(s)

Have the following object relation(s)

Type

Category

ALL

ALL

Only find IOCs flagged as to IDS

Exclude IOCs spotted in a warning list

First seen and Last seen

Attributes not having first seen or last seen set might not appear in the search

First seen date

Last seen date

First seen time

Last seen time

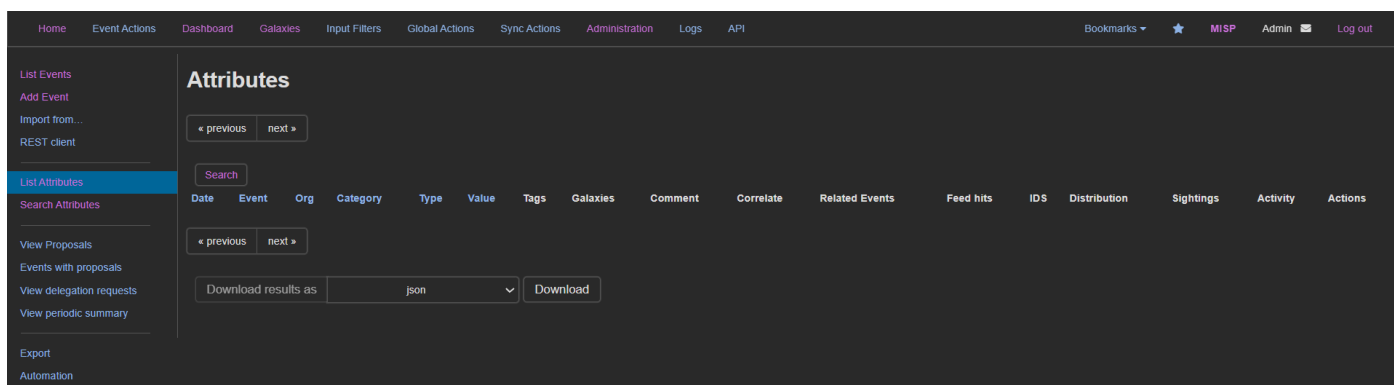
HH:MM:SS.ssssss+TT:TT

HH:MM:SS.ssssss+TT:TT

Expected format: HH:MM:SS.ssssss+TT:TT

Expected format: HH:MM:SS.ssssss+TT:TT

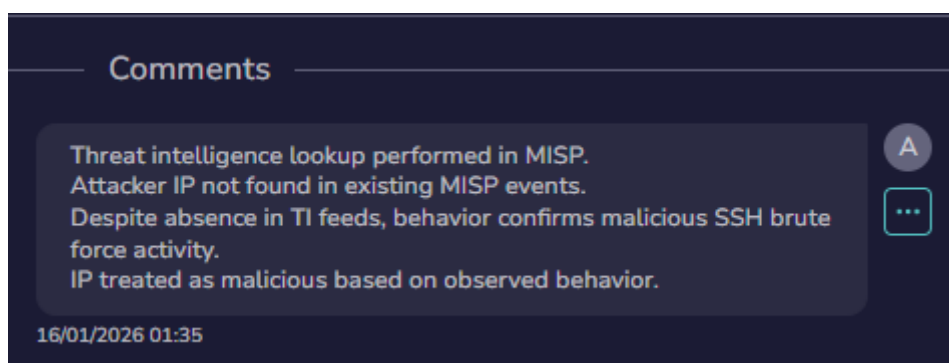
Search



◆ Step 10: Threat Intelligence Result Documented in TheHive

The outcome of the MISP lookup was documented in TheHive as a case comment.

Despite no prior threat intelligence findings, the IP was treated as malicious based on confirmed SSH brute force behavior.



◆ Step 11: Mitigation Actions on Target Host

Immediate mitigation actions were applied:

- Attacker IP blocked using firewall rules
- Fail2Ban enabled to prevent future brute force attempts
- SSH configuration hardened (root login and password authentication restricted)

These steps ensured attack containment.

```

ubuntu@ip-10-0-10-0:~$ sudo iptables -A INPUT -s 102.208.186.12 -j DROP
ubuntu@ip-10-0-10-0:~$ sudo iptables -L -n --line-numbers
Chain INPUT (policy ACCEPT)
num target prot opt source destination
1 DROP 0 -- 0.0.0.0/0 0.0.0.0/0
2 DROP 0 -- 0.0.0.0/0 0.0.0.0/0
3 DROP 0 -- 0.0.0.0/0 0.0.0.0/0
4 DROP 0 -- 102.208.186.12 0.0.0.0/0
5 DROP 0 -- 102.208.186.12 0.0.0.0/0

Chain FORWARD (policy ACCEPT)
num target prot opt source destination

Chain OUTPUT (policy ACCEPT)
num target prot opt source destination
ubuntu@ip-10-0-10-0:~$ sudo systemctl enable fail2ban
sudo systemctl start fail2ban
Synchronizing state of fail2ban.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable fail2ban
ubuntu@ip-10-0-10-0:~$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed: 0
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd
`- Actions
   |- Currently banned: 0
   |- Total banned: 0
   `-- Banned IP list:
ubuntu@ip-10-0-10-0:~$ sudo nano /etc/ssh/sshd_config
ubuntu@ip-10-0-10-0:~$ sudo systemctl restart ssh
ubuntu@ip-10-0-10-0:~$

# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
PermitEmptyPasswords no

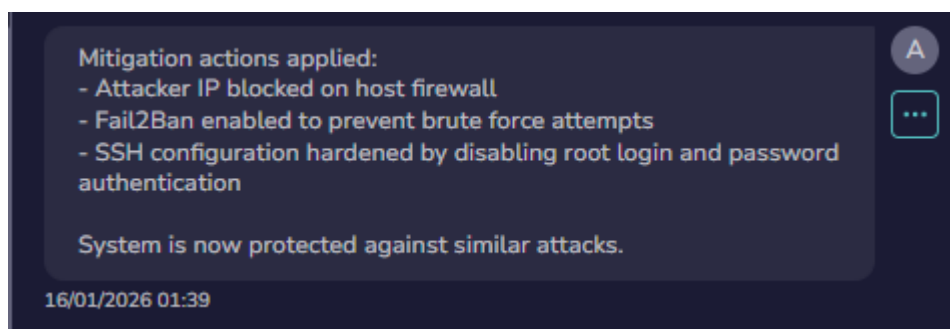
# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)

[ line 65/13
^G Help ^C Write Out ^W Where Is ^K Cut ^T Exec
^X Exit ^R Read File ^\ Replace ^U Paste ^J Just

```

◆ Step 12: Mitigation Comment Added in TheHive

All mitigation actions were documented within the case to maintain a full incident response record.



◆ Step 13: Case Tagged

Relevant case tags were added in TheHive to categorize the incident, including attack type and affected technology.

Tagging ensured proper classification and future searchability of the incident.

The screenshot shows the 'General' tab of a case in TheHive. The case title is 'Confirmed SSH Brute Force Attack - Linux Host'. The left sidebar shows the case ID as ~88383552, created by 'abdulrehmanthehivesoc', and created/updated on 16/01/2026. The main content area shows the title, tags, and a description field. The tags include 'credential-access', 'ssh', 'rule=100300', 'bruteforce', 'agent_ip=no agent ip', 'linux', 'wazuh', 'agent_name=ip', and 'agent_id=000'. The severity is set to 'MEDIUM' and the TLP/PAP levels are 'AMBER'.

#2 Confirmed SSH Brute Force Attack – Linux Host

id ~88383552
Created by abdulrehmanthehivesoc
Created at 16/01/2026 01:27
Updated at 16/01/2026 02:00

SEVERITY:MEDIUM
TLP:AMBER PAP:AMBER

General Tasks (0) Observables (2) TTPs (1) Attachments Timeline Report Pages History

* Title
Confirmed SSH Brute Force Attack – Linux Host

Tags
credential-access ssh rule=100300 bruteforce agent_ip=no agent ip linux wazuh agent_name=ip agent_id=000

Description

◆ Step 14: Case Closed as True Positive

The investigation was finalized by marking the case as **True Positive** and setting the status to **Resolved**.

A closing summary was added confirming successful detection, mitigation, and no further malicious activity.

The screenshot shows the 'Close case #2' dialog in TheHive. It has a title bar with a close button. The 'Status' is set to 'True positive'. The 'Impact' is set to 'No'. The 'Summary' field contains a text area with the following content: 'Confidence: High', 'This case involved a confirmed SSH brute force attack.', 'Detection, validation, threat intelligence enrichment, mitigation, and documentation were completed.', 'No further malicious activity observed after mitigation.', 'Case closed.', and 'Status: Resolved'. There is a 'Preview' button next to the summary text area.

Close case #2

* Status
True positive

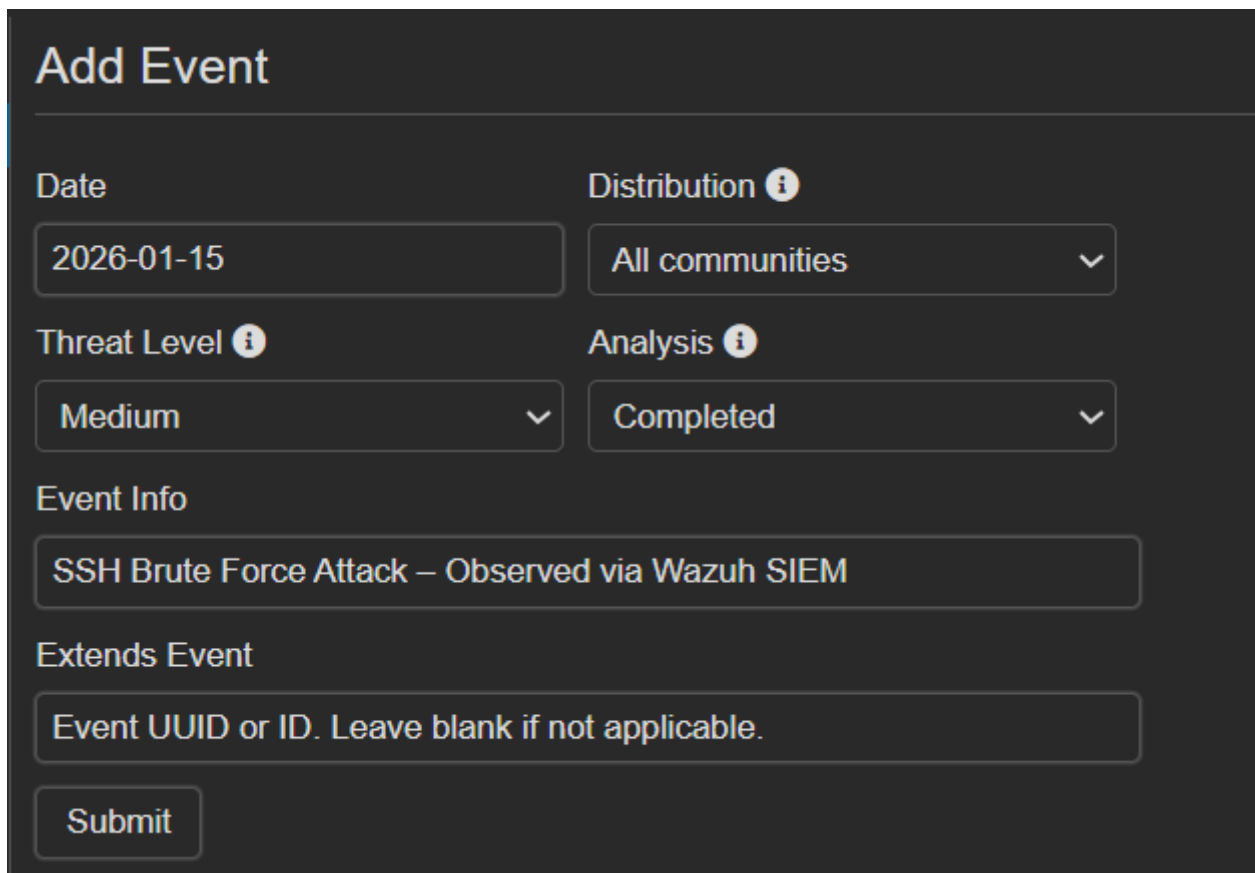
* Impact
☐ Yes ☒ No

* Summary
Confidence: High
This case involved a confirmed SSH brute force attack.
Detection, validation, threat intelligence enrichment, mitigation, and documentation were completed.
No further malicious activity observed after mitigation.
Case closed.
Status: Resolved

Preview

◆ Step 15: Threat Intelligence Sharing in MISP

A new MISP event titled “**SSH Brute Force Attack – Observed via Wazuh SIEM**” was created. Distribution was set to **All Communities** to share the IOC publicly.



Add Event

Date: 2026-01-15

Distribution ⓘ: All communities ▼

Threat Level ⓘ: Medium ▼

Analysis ⓘ: Completed ▼

Event Info: SSH Brute Force Attack – Observed via Wazuh SIEM

Extends Event: Event UUID or ID. Leave blank if not applicable.

Submit

◆ Step 16: Attribute Added to MISP Event

The malicious IP was added as:

- Category: Network activity
- Type: ip-src
- Contextual comment included
- First seen / last seen timestamps populated

Add Attribute

Category ⓘ
Type ⓘ

Network activity ▼
ip-src ▼

Distribution ⓘ

Inherit event ▼

Value

102.208.186.12

Contextual Comment

Confirmed SSH brute force source IP observed via Wazuh

☐ Batch import ⓘ
☐ For Intrusion Detection System
☐ Disable Correlation

First seen date 📅
Last seen date 📅

2026-01-16
2026-01-16

First seen time ⌚
Last seen time ⌚

2026-01-16 00:50:00.000000+00:00
2026-01-16 00:52:00.000000+00:00

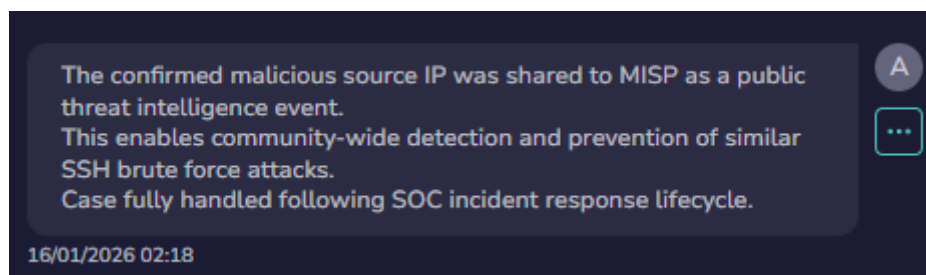
⌋ Expected format: HH:MM:SS.ssssss+TT:TT
⌋ Expected format: HH:MM:SS.ssssss+TT:TT

Submit
Cancel

◆ Step 17: Tags and MITRE Mapping in MISP

Standard MISP tags were added, including:

- TLP:AMBER
- MITRE ATT&CK Brute Force (T1110)



CONCLUSION

This project demonstrates the handling of a **real, live cyberattack** using industry-standard SOC tools and methodologies.

The incident was:

- Detected in real time
- Properly investigated
- Mitigated at the system level
- Documented in a case management platform
- Shared as threat intelligence for community benefit

SKILLS DEMONSTRATED

- SIEM alert triage and validation
- SOC case management (TheHive)
- Log analysis and threat confirmation
- MITRE ATT&CK mapping
- Incident response and mitigation
- Linux security hardening
- Threat intelligence enrichment and sharing (MISP)
- Professional documentation

FINAL NOTE

This was **not a simulated lab**.

This was a **real brute force attack**, handled end-to-end using SOC best practices.
